Supplementary material

Supplementary Table 1 Cochrane search strategy and keywords used for the search PICO

PICO Strategy	Participants		Intervention		Comparator		Outcome
Study Focus	Adult patients who require anticoagulation in the setting of a clinical trial		Direct-acting oral anticoagulant		Warfarin		Gastrointestinal bleeding*
Free text and MeSH terms (BOOLEAN operators to maximize yield)	Deep vein (venous) thrombosis (OR) Pulmonary embolism (OR) Thromboembolism (OR) Atrial fibrillation (OR) Prosthetic valve (AND) Clinical Trial	(AND)	Apixaban (OR) Rivaroxaban (OR) Dabigatran (OR) Edoxaban (OR) Betrixaban (OR) Oral anticoagulation (OR) Direct factor Xa Inhibitor (AND) Clinical Trial	(AND)	Warfarin (OR) Coumadin (OR) Acenocoumarol (OR) Vitamin K antagonists (AND) Clinical Trial	(AND)	*Outcome was not included to keep the search criteria broad-based on the assumption that gastrointestinal bleeding as a complication does not always get included in the title or abstract

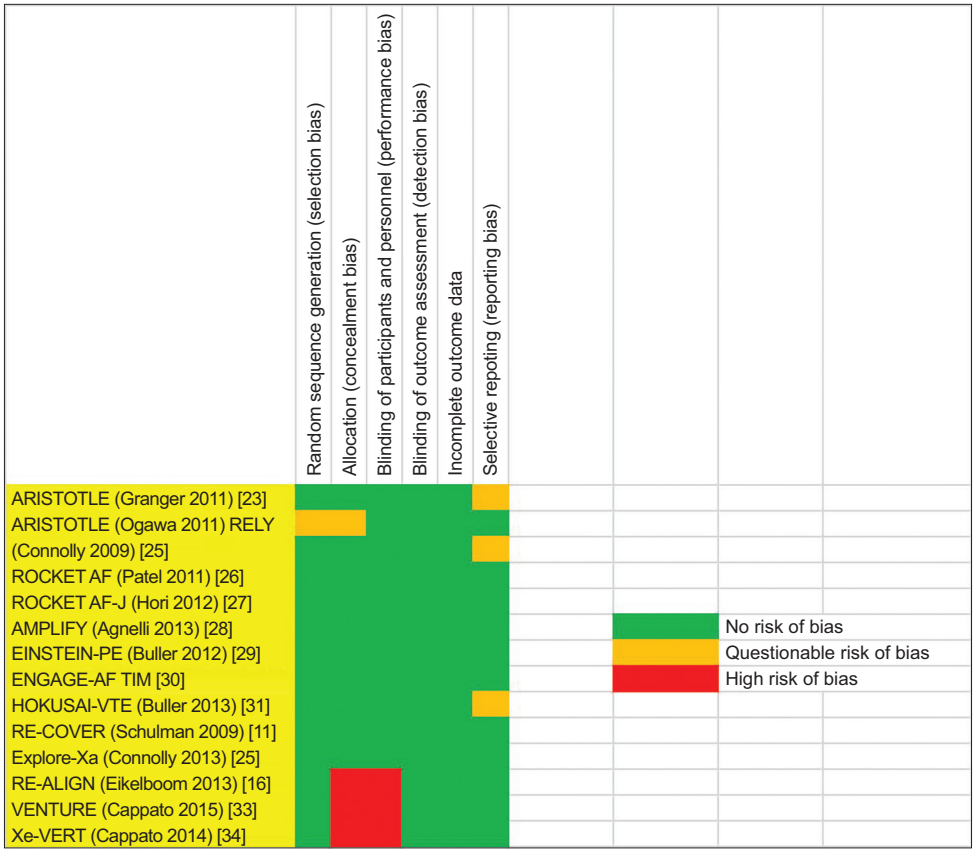
Participants = adult patients who require anticoagulation in the setting of a clinical trial

Intervention = Direct acting oral anticoagulant

Comparator = Warfarin

Outcome = Gastrointestinal bleeding

Search strategy

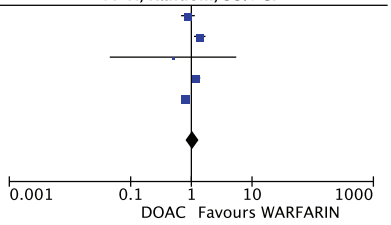


Supplementary Figure 1 Cochrane method for analysis of study quality

A

Study or Subgroup	DOAC Events	DOAC Total	WARFARIN Events	WARFARIN Total	Weight	Risk Ratio M-H, Random, 95% CI
AMPLIFY (Agnelli *et al* 2013) [28]	105	9120	119	9081	22.3%	0.88 [0.68, 1.14]
EINSTEIN PE (Buller *et al* 2012) [29]	190	7131	138	7133	24.1%	1.38 [1.11, 1.71]
HOKUSAI VTE (Buller *et al* 2013) [31]	1	2419	2	2413	1.0%	0.50 [0.05, 5.50]
RE ALIGN (Eikelboom *et al* 2013) [26]	368	11406	192	7036	25.9%	1.18 [1.00, 1.40]
RE-COVER (Schulman *et al* 2009) [11]	298	4112	368	4118	26.8%	0.81 [0.70, 0.94]
Total (95% CI)		**34188**		**29781**	**100.0%**	**1.03 [0.80, 1.32]**
Total events	962		819			

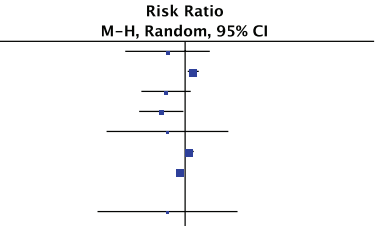
Heterogeneity: $Tau^2 = 0.05$; $Chi^2 = 21.15$, df = 4 (P = 0.0003); $I^2 = 81\%$
Test for overall effect: Z = 0.22 (P = 0.82)



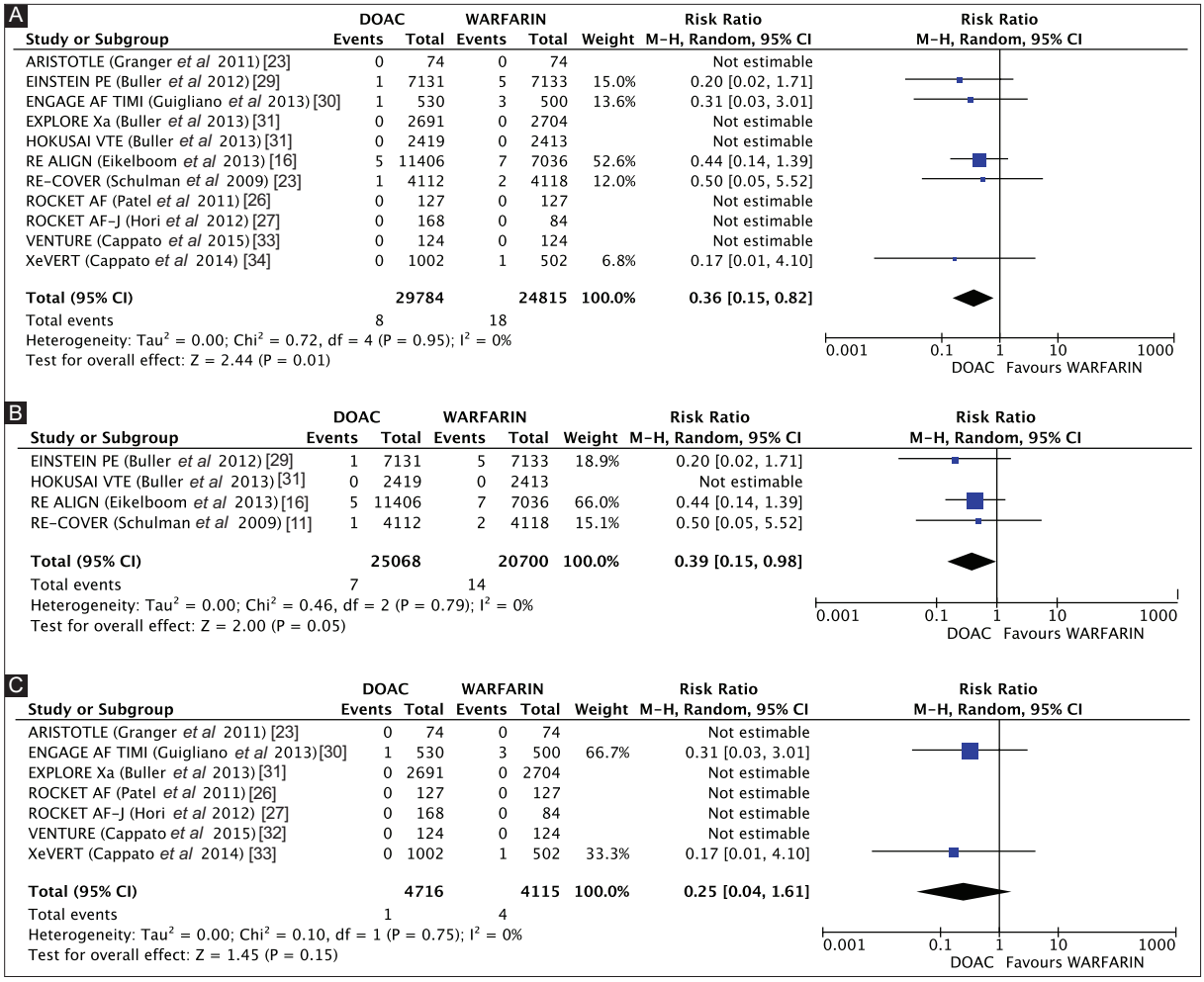
B

Study or Subgroup	DOAC Events	DOAC Total	WARFARIN Events	WARFARIN Total	Weight	Risk Ratio M-H, Random, 95% CI
ARISTOTLE (Granger *et al* 2011) [23]	2	74	4	74	2.6%	0.50 [0.09, 2.65]
EINSTEIN PE (Buller *et al* 2012) [29]	190	7131	138	7133	24.8%	1.38 [1.11, 1.71]
ENGAGE AF TIMI (Guigliano *et al* 2013) [30]	6	530	12	500	6.6%	0.47 [0.18, 1.25]
EXPLORE Xa (Buller *et al* 2013) [31]	7	2691	18	2704	7.8%	0.39 [0.16, 0.93]
HOKUSAI VTE (Buller *et al* 2013) [31]	1	2419	2	2413	1.3%	0.50 [0.05, 5.50]
RE ALIGN (Eikelboom *et al* 2013) [16]	368	11406	192	7036	26.2%	1.18 [1.00, 1.40]
RE-COVER (Schulman *et al* 2009) [11]	298	4112	368	4118	26.9%	0.81 [0.70, 0.94]
ROCKET AF (Patel *et al* 2011) [26]	0	127	0	127		Not estimable
ROCKET AF-J (Hori *et al* 2012) [27]	1	168	1	84	1.0%	0.50 [0.03, 7.90]
VENTURE (Cappato *et al* 2015) [33]	2	124	1	124	1.4%	2.00 [0.18, 21.77]
XeVERT (Cappato *et al* 2014) [34]	3	1002	1	502	1.5%	1.50 [0.16, 14.41]
Total (95% CI)		**29784**		**24815**	**100.0%**	**0.93 [0.70, 1.23]**
Total events	878		737			

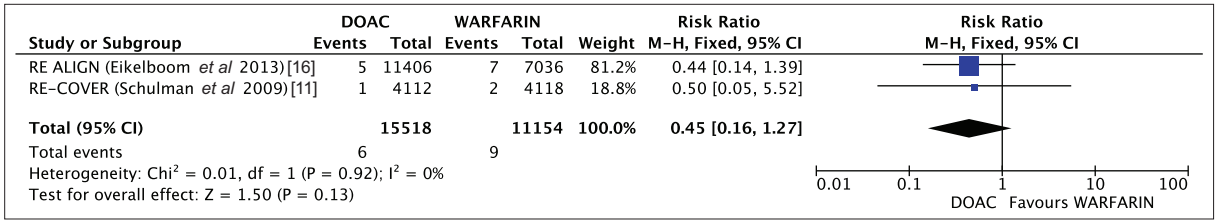
Heterogeneity: $Tau^2 = 0.07$; $Chi^2 = 28.32$, df = 9 (P = 0.0008); $I^2 = 68\%$
Test for overall effect: Z = 0.52 (P = 0.60)



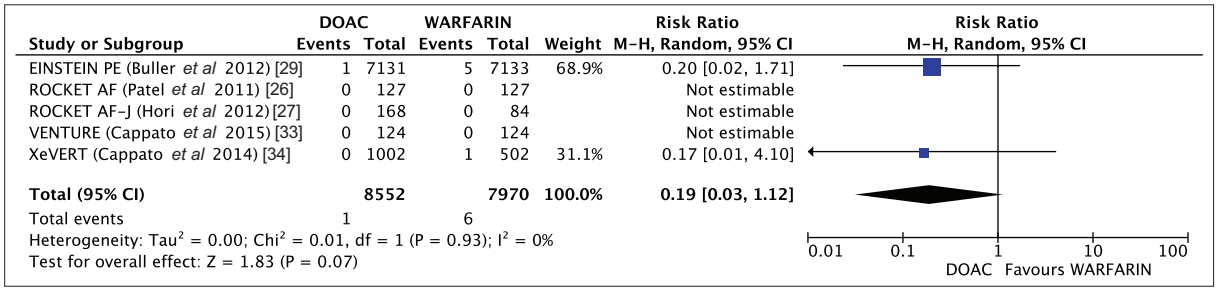
Supplementary Figure 2 (A) GI bleed < 60y; (B). GI bleed > 60y
DOAC, direct-acting oral anticoagulant; CI, confidence interval



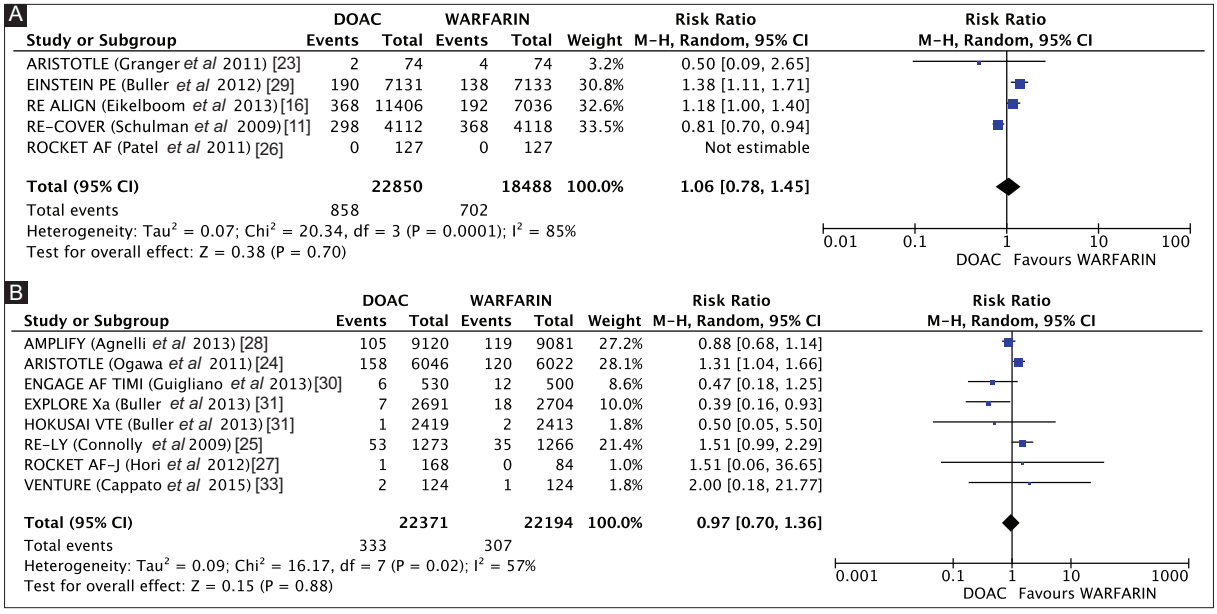
Supplementary Figure 3 (A) Any fatal GI bleed, (B) Fatal GI bleed <60y, (C) Fatal GI bleed > 60y
DOAC, direct-acting oral anticoagulant; CI, confidence interval; GI, gastrointestinal



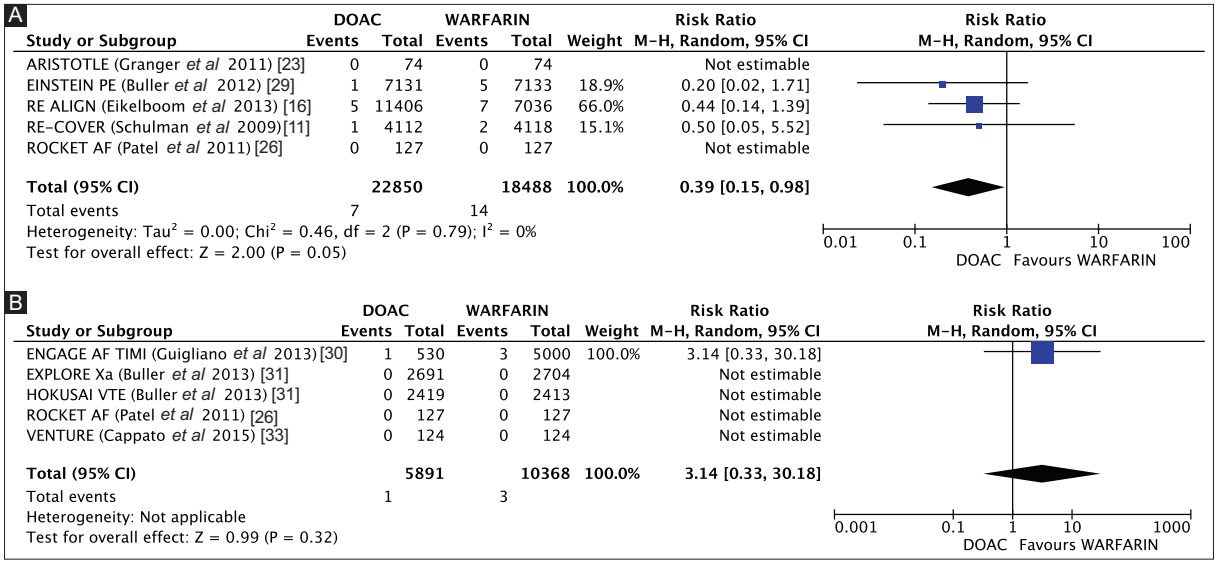
Supplementary Figure 4 Dabigatran fatal GI bleed
DOAC, direct-acting oral anticoagulant; CI, confidence interval



Supplementary Figure 5 Rivaroxaban fatal GI bleed
DOAC, direct-acting oral anticoagulant; CI, confidence interval; GI, gastrointestinal



Supplementary Figure 6 (A) Any GI bleed (INR<60% target therapeutic range) (B) Any GI bleed (INR>60% target therapeutic range)
DOAC, direct-acting oral anticoagulant; CI, confidence interval; GI, gastrointestinal; INR, international normalized ratio



Supplementary Figure 7 (A) Fatal GI bleed (INR < 60% target therapeutic range) (B) Fatal GI bleed (INR > 60% target therapeutic range)
DOAC, direct-acting oral anticoagulant; CI, confidence interval; GI, gastrointestinal; INR, international normalized ratio